COMS 3003A
Deferred Test 1

17 April, 2024

This is a closed book test.
Time allowance: 45 minutes

1. **(10 points)** We want to encode words in the decimal alphabet containing digits 0 through 9 in binary, making sure that the binary representation of a decimal word is as short as possible.
   (a) Explain in detail how we could do this.
   (b) How long would be the binary representation of a string 4998793421?

2. **(10 points)** Design a Turing machine that checks if the input binary string is a palindrome.

3. **(15 points)**
   (a) **(5 points)** Prove that every Turing machine M whose program contains instructions of the form
   $$q_i s_k \quad \rightarrow \quad q_j s_l S,$$
   where S stands for 'stay in place', can be converted to a Turing machine M' that can move right and left only once and solves the same problem. Assume that Turing machines have a one-way infinite tape, that all inputs are binary, and that the only take symbol that is not an input symbol is the blank symbol $\sqcup$.
   (b) **(10 points)** Give a precise (not big-Oh!) upper bound on the number of states of M'.

4. **(10 points)** Explain how a two-way infinite tape Turing machine can be can simulated by a one-way infinite tape Turing machine. You may assume the existence of any Turing machine mentioned in HW questions.